**Learner Full Name:** _____

**School:** _____ **Class:** _____

**Workstation Device:** _____ **Station #:** _____

# Session 11: Functions, Ranges & Visual Charts

In Session 10, you multiplied single cells together using =B2*C2. But what if your school store sells 100 different items? Typing =D2+D3+D4+D5...+D101 would take 20 minutes! Today, you will learn how to command the spreadsheet to calculate entire columns in a single breath using **Functions** and **Ranges**.

## 1. What is a "Range"? (The Power of the Colon :)

A **Range** is a block of connected cells. In spreadsheet language, we write a range using the first cell address, a **colon (:)**, and the last cell address.

The colon simply means **"THROUGH TO"**:




*Figure 11.0: Anatomy of the SUM function. The colon (:) tells the computer to add every cell from D2 to D5.*

## 2. Formula vs. Function: What is the Difference?

### FORMULA (Manual Math)

A math equation you write by hand using basic symbols: =B2*C2 or =A1+A2+A3.

Best for simple calculations between 2 or 3 cells.

### FUNCTION (Preset Magic Word)

A built-in preset shortcut word provided by the spreadsheet to crunch massive lists of numbers: =SUM(D2:D5).

Best for adding or averaging 5, 50, or 5,000 cells at once!

# THE "CORE 4" STATISTICAL FUNCTIONS

## 1. Meet the Big Four Built-in Functions

In school, business, and science, these four functions do 90% of all data work:

| Function Name | What It Does | How to Write It in a Cell |
|---|---|---|
| **=SUM(...)** | **Adds** every single number in the range together to calculate a Grand Total. | =SUM(D2:D5) |
| **=AVERAGE(...)** | Adds the numbers and divides by the count to find the **mean (typical) score**. | =AVERAGE(C2:C5) |
| **=MAX(...)** | Scans the entire list and displays the **highest (maximum)** value. | =MAX(B2:B5) |
| **=MIN(...)** | Scans the entire list and displays the **lowest (minimum)** value. | =MIN(C2:C5) |

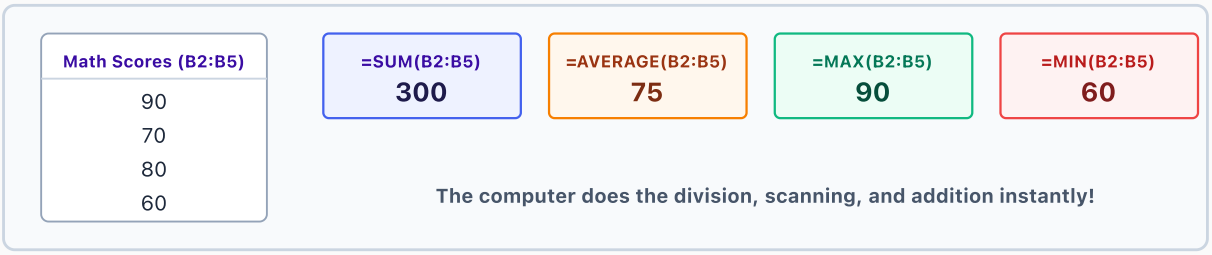



*Figure 11.1: The Core 4 functions operating on the exact same range of exam scores.*

## 2. Check Your Thinking

Look at the numbers: 10, 20, 30, 40 located in cells A1:A4.

**A.** What is the result of =SUM(A1:A4)? _____

**B.** What is the result of =MAX(A1:A4)? _____

**C.** What is the result of =AVERAGE(A1:A4)? _____

## 1. Why Do We Use Charts?

Human beings are visual thinkers. When a chief, headteacher, or market board inspects a spreadsheet with 500 rows of raw numbers, their eyes get tired. But when you convert that data into a **Chart**, patterns, trends, and comparisons leap out in **one second**!

## 2. The Three Classic School Chart Types

### 📊 COLUMN / BAR CHART

Uses vertical or horizontal bars to **compare separate categories** side by side.

*Best for:* Comparing sales of different items, attendance across classrooms, or harvest yields per village.

### 🥧 PIE CHART (Slices of a Cake)

A circle sliced into wedges showing **parts of a 100% whole**.

*Best for:* School budget spending (food, books, sports) or waste composition (plastics vs. organic food scraps).

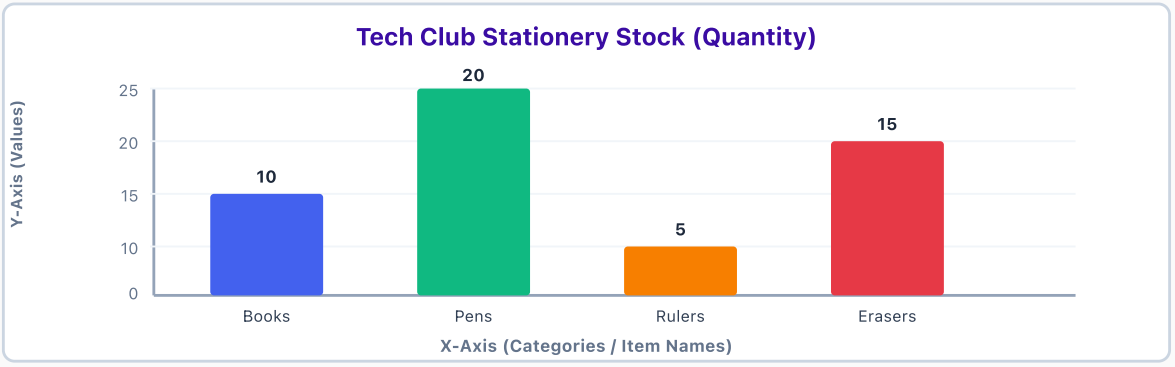



*Figure 11.2: Anatomy of a Column Chart. Data Labels on top of each bar display exact quantities.*

# THE GOLDEN RULE OF CHART SELECTION

## 1. What Cells Do I Highlight Before Making a Chart?

A chart is only as smart as the cells you feed into it. The computer does not guess: you must highlight the exact labels and numbers you want to visualize.

> 🧯 **THE #1 BEGINNER CHART MISTAKE:**
>
> **NEVER INCLUDE THE "GRAND TOTAL" ROW IN YOUR CHART!**
>
> If you include the Total row, the Total bar will be huge (e.g. 50 items) and will dwarf all your individual products, making them look like tiny flat slivers!




*Figure 11.3: Data selection rule for charts. Always select the headers and items, but stop before the Total row!*

## 2. Troubleshooting Chart Issues

| Problem Observed | Why It Happened | Immediate Fix |
|---|---|---|
| **My chart is completely blank!** | You highlighted empty white cells before clicking Insert Chart. | Delete chart ( `Del` ), highlight your actual data table, and re-insert. |
| **The bars have no labels under them.** | You only selected Column B (numbers) and forgot to select Column A (item names). | Select **both** columns: from A1 across to B5! |

# GUIDED HANDS-ON LAB: TOTALS & CHARTS

## 1. Lab Mission: Add Grand Totals, Stats & a Visual Chart

Follow these 6 steps inside your existing spreadsheet file from Session 10:

**1** **Open Your File:** Launch Excel, Calc, or Sheets → Open Stationery_Store_Ama.xlsx from your BIH_Practice folder.

**2** **Calculate Grand Total using =SUM:**
- Click inside Cell **A6**, type: GRAND TOTAL.
- Click inside Cell **D6**, type: =SUM(D2:D5) and tap `Enter`.

*Observation:* D6 displays **162.50**!

**3** **Calculate Statistics:**
- In Cell **A7**, type: Average Price → in **C7**, type: =AVERAGE(C2:C5)
- In Cell **A8**, type: Highest Price → in **C8**, type: =MAX(C2:C5)
- In Cell **A9**, type: Lowest Price → in **C9**, type: =MIN(C2:C5)

**4** **Select Data for the Chart:**
Click in Cell **A1**, hold down the left mouse button, and drag across and down to Cell **B5**.
*Check:* Cells A1 through B5 are highlighted blue/green. (Notice you left Row 6 OUT!).

**5** **Insert the Chart:**
Click the **Insert** tab on the Ribbon → click **Column / Bar Chart** → select **2D Clustered Column**.
A colorful bar chart appears right on your sheet!

**6** **Add Title & Test Dynamic Magic:**
Double-click the default chart title and rename it: Stationery Stock Quantities.
Now change Cell **B2 (Books)** from 10 to **35**.
*Watch:* The Grand Total updates AND the blue chart bar grows taller instantly! Save: `Ctrl + S`.

## 2. Visual Target (What Your Finished Sheet Should Look Like)

| 1 | Item Description | Quantity | Price (GHS) | Line Total |
|---|---|---|---|---|
| 2 | Exercise Books | 10 | 8.00 | 80.00 |
| 3 | Ballpoint Pens | 20 | 2.00 | 40.00 |
| 4 | Wooden Rulers | 5 | 4.00 | 20.00 |
| 5 | Pencil Erasers | 15 | 1.50 | 22.50 |
| 6 | **GRAND TOTAL** | | | **162.50** |
| 7 | Average Price | | **3.88** | |
| 8 | Highest Price | | **8.00** | |
| 9 | Lowest Price | | **1.50** | |

# SESSION 11 REVIEW & VERIFICATION



## 1. Match the Function to its Job

Draw a line connecting each spreadsheet function to its correct mathematical job:

| | |
|---|---|
| **=SUM(B2:B10)** | • Finds the smallest / lowest value in the range. |
| **=AVERAGE(B2:B10)** | • Adds up all numbers in the range to give a grand total. |
| **=MAX(B2:B10)** | • Represents a range of cells from D2 through to D20. |
| **=MIN(B2:B10)** | • Finds the largest / highest value in the range. |
| **D2:D20** | • Calculates the typical mean score of the range. |

## 2. Written Review Questions

**Question 1:** Why should you NEVER include the "Grand Total" row when highlighting data to build a Column chart?

**Question 2:** What does the colon symbol (**:**) in the formula =SUM(A1:A50) stand for?

**Question 3:** Which type of chart is shaped like a circular cake and is best for showing parts of a 100% whole budget?

## 3. Self-Assessment & Mentor Sign-Off

Tick the column that reflects your confidence at the end of Level 5 (Spreadsheets):

| Spreadsheet Functions & Charting Competency | With Help | Almost Alone | All By Myself! |
|---|---|---|---|
| 1. I can write a range of cells using colon notation (e.g. D2:D5). | ○ | ○ | ○ |
| 2. I can calculate grand totals using =SUM and averages using =AVERAGE. | ○ | ○ | ○ |
| 3. I can find the highest and lowest numbers using =MAX and =MIN. | ○ | ○ | ○ |
| 4. I can select items and quantities (leaving out Total) to create a Column Chart. | ○ | ○ | ○ |

**Mentor Signature:** _____ **Date Completed:** _____